

Carr-Madan & VIX

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Carr-Madan & VIX Derivation

1. Log Dynamics and Ito's Lemma

Under the risk-neutral measure: $dS_t = rS_t dt + \sigma_t S_t dW_t$

Applying Ito's lemma to $\log S_t$: $d \log S_t = \frac{1}{S_t} dS_t - \frac{1}{2} \frac{1}{S_t^2} (dS_t)^2$

Since $(dS_t)^2 = \sigma_t^2 S_t^2 dt$, $d \log S_t = \left(r - \frac{1}{2} \sigma_t^2\right) dt + \sigma_t dW_t$

Integrating from 0 to T ,

$$\log S_T - \log S_0 = \int_0^T \left(r - \frac{1}{2} \sigma_t^2\right) dt + \int_0^T \sigma_t dW_t$$

Taking expectations,

$$\mathbb{E}[\log S_T] = \log S_0 + rT - \frac{1}{2} \mathbb{E} \left[\int_0^T \sigma_t^2 dt \right]$$

Then,

$$\mathbb{E} \left[\int_0^T \sigma_t^2 dt \right] = 2(rT - \mathbb{E}[\log(S_T/S_0)])$$

Let $F = S_0 e^{rT}$, then:

$$\mathbb{E} \left[\int_0^T \sigma_t^2 dt \right] = -2 \mathbb{E} \left[\log \left(\frac{S_T}{F} \right) \right]$$

2. Carr-Madan Spanning Formula

For a twice differentiable payoff f , we have:

$$f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)(K - S_T)^+ dK + \int_F^\infty f''(K)(S_T - K)^+ dK$$

Choosing $f(S_T) = -2 \log \left(\frac{S_T}{F} \right)$

Then, $f'(S) = -\frac{2}{S}$, $f''(S) = \frac{2}{S^2}$

So $f''(K) = \frac{2}{K^2}$

Substitute into the spanning formula:

$$f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F \frac{2}{K^2} (K - S_T)^+ dK + \int_F^\infty \frac{2}{K^2} (S_T - K)^+ dK$$

3. Taking Expectations and Vanishing of Linear Terms

From Carr-Madan:

$$f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F \frac{2}{K^2} (K - S_T)^+ dK + \int_F^\infty \frac{2}{K^2} (S_T - K)^+ dK$$

Since $f(S) = -2 \log\left(\frac{S}{F}\right)$, $f(F) = -2 \log(1) = 0$

Similarly, $f'(S) = -\frac{2}{S} \Rightarrow f'(F) = -\frac{2}{F}$

Thus the linear component becomes:

$$f(F) + f'(F)(S_T - F) = -\frac{2}{F}(S_T - F)$$

Now take expectations under the risk-neutral measure:

$$\mathbb{E}\left[-\frac{2}{F}(S_T - F)\right] = -\frac{2}{F}(\mathbb{E}[S_T] - F)$$

By definition $\mathbb{E}[S_T] = F$, so $\mathbb{E}[f(F) + f'(F)(S_T - F)] = 0$

Hence, linear terms vanish in expectation

Taking expectations of the full decomposition:

$$\mathbb{E}[f(S_T)] = \int_0^F \frac{2}{K^2} P(K) dK + \int_F^\infty \frac{2}{K^2} C(K) dK$$

From Section 1:

$$\mathbb{E}[f(S_T)] = \mathbb{E}\left[\int_0^T \sigma_t^2 dt\right]$$

Thus:

$$\mathbb{E}\left[\int_0^T \sigma_t^2 dt\right] = \int_0^F \frac{2}{K^2} P(K) dK + \int_F^\infty \frac{2}{K^2} C(K) dK$$

4. VIX Formula

The VIX (CBOE) discretizes this integral:

$$\text{VIX}^2 \approx \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} Q(K_i)$$

where:

- $Q(K_i)$ are out-of-the-money option prices
- both puts ($K_i < F$) and calls ($K_i > F$) are used
- ΔK_i are strike spacings

5. Equivalence Between Carr-Madan and VIX

From Carr-Madan,

$$\mathbb{E} \left[\int_0^T \sigma_t^2 dt \right] = \int_0^F \frac{2}{K^2} P(K) dK + \int_F^\infty \frac{2}{K^2} C(K) dK$$

Define a unified function of out-of-the-money option prices:

$$Q(K) = \begin{cases} P(K), & K < F \\ C(K), & K > F \end{cases}$$

Then the expression becomes

$$\mathbb{E} \left[\int_0^T \sigma_t^2 dt \right] = \int_0^\infty \frac{2}{K^2} Q(K) dK$$

Using a discrete grid of strikes $\{K_i\}_{i=1}^N$,

$$\Delta K_i = \frac{K_{i+1} - K_{i-1}}{2}$$

Using the Riemann sum approximation,

$$\int_0^\infty \frac{2}{K^2} Q(K) dK \approx \sum_{i=1}^N \frac{2}{K_i^2} Q(K_i) \Delta K_i$$

Thus,

$$\mathbb{E} \left[\int_0^T \sigma_t^2 dt \right] \approx \sum_{i=1}^N \frac{2 \Delta K_i}{K_i^2} Q(K_i)$$

Finally, dividing by T (variance per unit time),

$$\frac{1}{T} \mathbb{E} \left[\int_0^T \sigma_t^2 dt \right] \approx \frac{2}{T} \sum_{i=1}^N \frac{\Delta K_i}{K_i^2} Q(K_i)$$

Which is identical to

$$\text{VIX}^2 \approx \frac{2}{T} \sum_{i=1}^N \frac{\Delta K_i}{K_i^2} Q(K_i)$$

Therefore

In the limit:

$$\text{VIX}^2 \rightarrow \frac{1}{T} \int_0^\infty \frac{2}{K^2} Q(K) dK$$

which is the Carr-Madan representation of expected variance.